

Jacob Davis

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Education

Clemson University

Clemson, SC

Bachelor of Science in Computer Science and Mathematical Sciences (GPA: 3.97)

Aug. 23 – May. 27

- **Clubs & Affiliations:** Clemson Honors College, Undergraduate Teaching Assistant, Clemson Symphony Orchestra (Leader)

Experience

Google

Sunnyvale, CA

Software Engineer Intern

May. 26 – Aug. 26

- Engineered an on-demand allocation optimization for low-priority miss queues, up to **doubling** theoretical cloud VM host capacity.
- Developed an asynchronous thread pool callback mechanism for queue expansion triggered by internet traffic, incorporating safe state restoration during hitless restarts and preventing global buffer pool exhaustion under network stress.
- **Skills Used:** C++, Software-Defined Networking (SDN), Distributed Systems, Concurrency, Packet Processing, Live Migration

Clemson Research Computing and Data

Clemson, SC

Software Engineer Intern

Jan. 26 – May. 26

- Optimized Clemson's OpenAI LLM service by reducing database reads with local caching, reducing website latency by **20%**.
- Architected a benchmark suite for LLM services under varying software and hardware configurations using a Kubernetes cluster.
- **Skills Used:** Go, Kubernetes, OpenAI API, Slurm, Shell Scripting, MySQL

NASA Goddard Space Flight Center

Greenbelt, MD

Software Engineer Intern

Jun. 25 – Jan. 26

- Implemented a sky localization algorithm for compact binary mergers detected by LIGO/Virgo/KAGRA using GPU programming, achieving a **performance increase by a factor of five** compared to C code parallelized on the CPU.
- Migrated performance-critical code in open source software used in production environments at gravitational-wave observatories to parallel processing frameworks, **reducing runtime by 99%** compared to sequential code.
- Presented in the 247th American Astronomical Society Meeting and Clemson's HPC Day, winning the **Most Impactful** poster award.
- **Skills Used:** JAX, CUDA, OpenCL, OpenMP, NumPy, Python, C

Clemson University International Center for Automotive Research

Greenville, SC

Software Engineer Intern

Aug. 24 – May. 25

- Trained and fine-tuned YOLOv8 and LiDAR segmentation models using a custom labeled dataset of outdoor environments, achieving **95% accuracy** in differentiating classes of trails and vegetation.
- Collaborated with Applied Research Associates to develop a sensor data collection system for an off-road autonomous vehicle.
- **Skills Used:** PyTorch, MATLAB, Classification Modeling, ZED SDK, Boson SDK

Cadence Design Systems

San Jose, CA

Application Engineer Intern

May. 24 – Aug. 24

- Achieved **99% functional coverage** in a sous vide controller design using Universal Verification Methodology, pinpointing over **20** bugs in the RTL code across **30+** features.
- Designed C++ and Python programs to automate analyzing and generating simulations, increasing workflow efficiency by **50%**.
- **Skills Used:** UVM, SystemVerilog, Linux, Cadence Xcelium, Cadence SimVision

Projects

High-Performance Cluster Computing | Spack, Slurm, OpenMPI, Docker, Linux

Jan. 25 – Present

- Assemble a four-node Raspberry Pi mini-cluster to simulate Clemson's **TOP500** Palmetto Cluster by configuring the environment using Spack to run OpenMPI applications and LINPACK benchmarks (maximum performance of **~30 GFLOPs**).
- Won **first place** in the interviews for a climate emulator at SC25's international Student Cluster Competition (SCC).

HPC Cluster System Admin Dashboard | Python, JavaScript, Vue, MongoDB, Vercel

Jan. 26

- Engineered a real-time file monitoring system for HPC clusters featuring an interactive dashboard and terminal navigator.
- Won the **"Best Use of MongoDB Award"** at HoyaHacks against **400+ participants**.

Computer Vision Paper Synthesizer | Python, YOLO, MediaPipe

Sep. 25

- Leveraged YOLOv8 and MediaPipe to draw and play a synthesizer on paper, achieving **90% accuracy** in detecting keys.
- Won the **"Best Musical Hack Award"** at HackMIT against **300+ teams**.

Asteroid Collision Simulator | JavaScript, Vue, Godot

Sep. 25

- Built an asteroid collision simulator showcasing seismic impact activity, meteor impact visualizations, and high user accessibility.
- Won the **"Most Inspirational Award"** at the NASA Space Apps Challenge's third largest event.